

Topic:-

Prepare a model of a cell using different eco-friendly materials.

Introduction:-

All living organisms on Earth—from tiny microscopic amoebas to giant banyan trees and human beings—are composed of microscopic building blocks called cells. The cell is rightly described as the structural and fundamental functional unit of life. English scientist Robert Hooke first discovered cells in 1665 when he observed a thin slice of bottle cork under a primitive microscope.

Inside a living cell, there is a jelly-like fluid called cytoplasm containing microscopic specialized functional units termed cell organelles. Each organelle carries out a specific duty to keep the cell alive—just like organs do in our human body. For school science fairs, students often construct cell models using non-biodegradable thermocol and toxic plastic clay, which harm our environment. In this project, we created a 3D model of a plant cell using completely eco-friendly, biodegradable, and recycled household materials.

❖ Main Parts of a Plant Cell:-

- **Cell Wall:** Outer rigid cellulose protective wall found only in plant cells.
- **Nucleus:** The master control center storing genetic instructions (DNA).
- **Mitochondria:** The cellular powerhouses generating energy molecules.
- **Chloroplasts:** Green solar food factories performing photosynthesis.

Aim & Eco-Friendly Materials:-

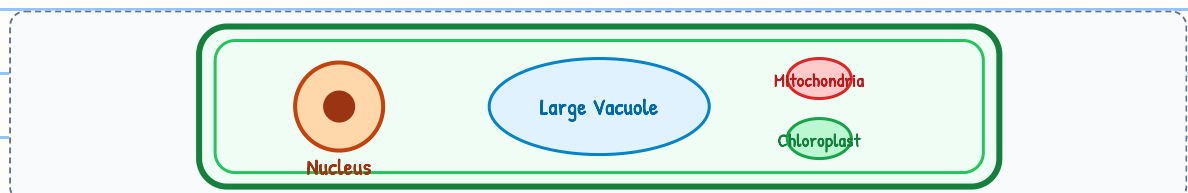
Aim: To design, build, and label a 3D physical model of a eukaryotic plant cell using completely natural, biodegradable, and recycled household materials.

❖ *Objectives of the Project:-*

- To physically represent plant cell organelles without using non-biodegradable thermocol.
- To understand the specific biological function of each organelle in a plant cell.
- To demonstrate that eco-friendly school crafts can be beautiful, durable, and green.

❖ *Natural Materials Selected:-*

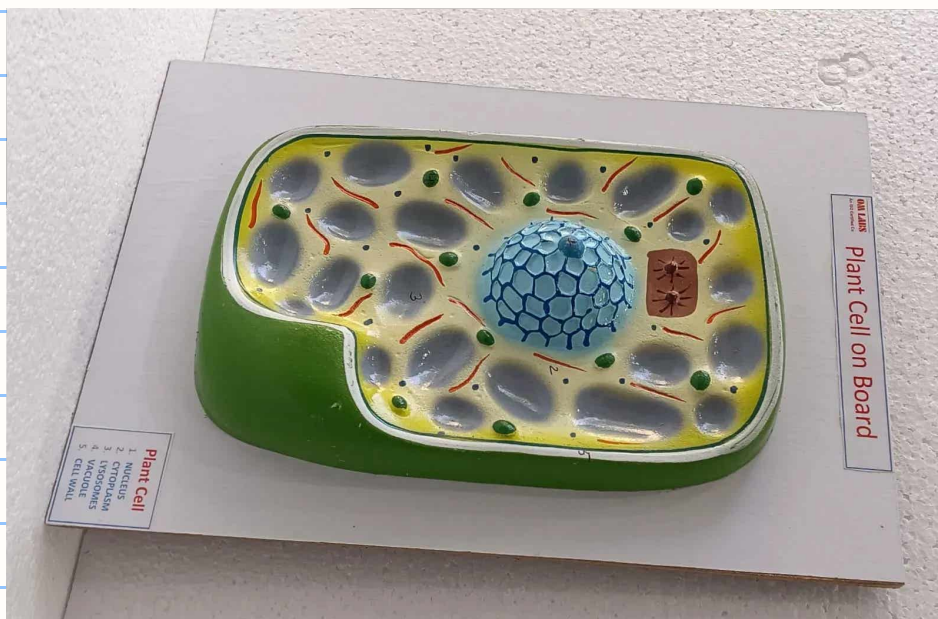
- Cardboard shoebox tray (base), thick jute rope (cell wall), cotton string (plasma membrane).
- Half coconut shell + walnut (nucleus), red rajma beans (mitochondria), green cardamom pods (chloroplasts), folded cardboard ribbons (Golgi and ER).



Step-by-Step Model Construction:-

We constructed our eco-friendly 3D cell model following these creative steps:

- **Step 1: Cell Base & Cytoplasm:** Took an empty sturdy cardboard box lid and lined it with pale green wheat flour dough to represent the watery cytoplasmic jelly.
- **Step 2: Boundary Layers:** Glued thick brown jute rope along the outer rim for the rigid cellulose *Cell Wall*. Stuck thin cotton twine immediately inside it for the flexible *Plasma Membrane*.
- **Step 3: Nucleus & Nucleolus:** Placed half a cut coconut shell towards one side to represent the nuclear envelope. Glued a dark walnut inside it to represent the dense *Nucleolus*.
- **Step 4: Energy & Food Organelles:** Glued 4 halved red kidney beans (Rajma) for *Mitochondria* and green cardamom pods (Elaichi) for *Chloroplasts*.
- **Step 5: Packaging & Storage:** Folded corrugated cardboard strips for the *Endoplasmic Reticulum* and made a large blue paper pouch for the *Central Vacuole*. Toothpick paper flags labeled each organelle clearly.



Organelle Roles & Eco-Material Table:-

The reference table below correlates the eco-friendly craft components with biological functions:

Cell Organelle	Natural Material Used	Biological Function in Plant Cell
Cell Wall	Thick brown jute rope	Provides rigid mechanical support, protection, and fixed shape
Cell Membrane	Fine white cotton twine	Selectively permeable barrier controlling entry/exit of nutrients
Nucleus & Nucleolus	Coconut shell + walnut	Controls all cellular activities, metabolism, and stores DNA
Mitochondria	Red kidney beans (Rajma)	Cellular powerhouse: produces energy molecules (ATP)
Chloroplasts	Green cardamom pods	Contains chlorophyll pigment to produce glucose via photosynthesis
Central Vacuole	Light blue folded paper pouch	Maintains cell turgidity; stores water, mineral salts, and wastes

Scientific Analysis & Discussion:-

Constructing this physical model reinforced key biological distinctions between plant and animal cells:

- **Why Plants Need a Rigid Cell Wall:** Unlike animals, plants cannot walk away to seek shelter during severe storms, heat, or heavy rains. The rigid cellulose cell wall withstands high internal osmotic water pressure without bursting.
- **Large Central Vacuole vs Small Vacuoles:** A mature plant cell possesses a giant central vacuole that occupies up to 90% of total cell volume. When full of water, it exerts turgor pressure against the cell wall, keeping plant stems upright. Animal cells have only tiny temporary vacuoles.
- **Chloroplasts as Solar Panels:** Chloroplasts contain stacks of thylakoid membranes (like stacks of coins) where green chlorophyll molecules trap solar photons to synthesize carbohydrates from CO₂ and water.

❖ Why Thermocol Should Be Banned in Schools:-

Thermocol is expanded polystyrene that takes over 500 years to decompose and emits toxic carcinogenic black smoke when burned. Natural jute, seeds, and cardboard decompose in garden soil within 6 weeks!

Advantages & Educational Impact:-

Our eco-friendly model delivered several significant advantages during school display:

- **1. Tactile Hands-On Learning:** Touching 3D textures (rough coconut shell, smooth seeds, twisted ropes) makes cell anatomy memorable for students compared to flat 2D textbook drawings.
- **2. Zero Environmental Footprint:** 100% biodegradable and assembled entirely from kitchen and recycling bin scraps, costing under ₹10.
- **3. Inspiring Classmates:** Visiting students from Classes 6 and 7 were inspired to stop purchasing plastic clay and thermocol for their science exhibits.

❖ Care and Preservation of Natural Models:-

- Spray with a light coat of natural neem oil spray to keep kitchen seeds free from ants.
- Store in a dry, ventilated display cupboard to prevent wheat dough from getting moldy.

What We Learned & Conclusion:-

- The cell is the basic structural and functional unit of life containing coordinated organelles.
- Plant cells possess distinctive features: a rigid cell wall, photosynthetic chloroplasts, and a large central vacuole.
- Each organelle performs a dedicated task: nucleus controls, mitochondria energizes, chloroplasts synthesize food.
- High-quality, durable 3D science models can be created sustainably using natural, eco-friendly materials.

Conclusion:-

Building a 3D plant cell model using eco-friendly materials successfully united cellular biology with environmental responsibility. The model demonstrated that science education can be engaging, visually captivating, and completely harmless to mother nature. The project enriched our whole class's understanding of life at the microscopic level while promoting sustainable, green living.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 10: Cell and Cell Organelles).
- NCERT Class 8 Science — Cell: Structure and Functions.
- Eco-Friendly Science Fair Project Handbook, Balbharti Pune.